

Accepted Manuscript

Platinum doping effect on In_2O_3 MSM IR photodetectors

Tarık Asar, Veysel Baran, Gürkan Kurtuluş, Meltem Dönmez, Süleyman Özçelik

PII: S0749-6036(18)30727-4

DOI: [10.1016/j.spmi.2018.05.035](https://doi.org/10.1016/j.spmi.2018.05.035)

Reference: YSPMI 5702

To appear in: *Superlattices and Microstructures*

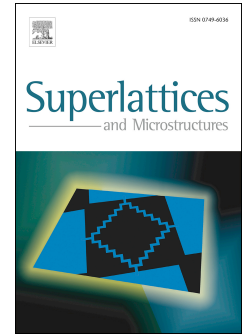
Received Date: 9 April 2018

Revised Date: 17 May 2018

Accepted Date: 18 May 2018

Please cite this article as: Tarı. Asar, V. Baran, Gü. Kurtuluş, M. Dönmez, Sü. Özçelik, Platinum doping effect on In_2O_3 MSM IR photodetectors, *Superlattices and Microstructures* (2018), doi: 10.1016/j.spmi.2018.05.035.

This is a PDF file of an unedited manuscript that has been accepted for publication. As a service to our customers we are providing this early version of the manuscript. The manuscript will undergo copyediting, typesetting, and review of the resulting proof before it is published in its final form. Please note that during the production process errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.



Platinum Doping Effect on In_2O_3 MSM IR Photodetectors

Tarık ASAR^{1,2,*}, Veysel BARAN^{1,2}, Gürkan KURTULUŞ², Meltem DÖNMEZ²
and Süleyman ÖZÇELİK^{1,2}

¹Department of Physics, Faculty of Science, Gazi University, 06500 Ankara, Turkey

²Photonics Application and Research Center, Gazi University, 06500 Ankara, Turkey

*Corresponding Author: trkasar@gazi.edu.tr

Abstract

The un-doped and platinum-doped In_2O_3 thin films were deposited for the investigation of platinum doping effect on the structural, morphological and electrical properties of In_2O_3 based metal-semiconductor-metal infrared photodetectors. The In_2O_3 metal-semiconductor-metal infrared photodetectors were also fabricated for obtaining the electrical characteristics such as the carrier recombination lifetime, diffusion length, carrier density, mobility, saturation current, barrier height, series resistance and ideality factor. It was seen from the characterization results that the *Pt* dopant usage has a prominent effect on properties of In_2O_3 metal-semiconductor-metal infrared photodetectors which are not extensively studied in the literature.

Keywords: In_2O_3 ; MSM; IR Photodetectors; XRD; HALL; I-V.

1. Introduction

Transparent conductive oxide (TCO) materials such as indium oxide (In_2O_3), tin-doped indium oxide (ITO), zirconium dioxide (ZrO_2), tungsten oxide (WO_3), zinc oxide (ZnO), aluminum-doped zinc oxide (AZO), titanium oxide (TiO_2) and gallium-doped zinc oxide (GZO) have been attracted noteworthy interest owing to their extensive applications in the optoelectronic and electrochemical areas [1–6]. Among these materials, In_2O_3 has been comprehensively used as an n-type semiconducting binary oxide material, which has wide band gap (3.5–3.7 eV), high electrical conductivity, high optical transparency and excellent carrier transport capability [7–9]. Despite these, the structural, electrical and optic properties of In_2O_3 thin films have been intensely debated [10]. It was seen that the properties of the TCO based devices can be enhanced with adjusting the nanostructured coatings [11,12]. Thereby, the enhancement on the properties of In_2O_3 thin film was obtained by doping the pure In_2O_3 with

small amount of *Pt* noble metal [13,14]. Other frequently usage of the doped indium oxide is also *ITO* which is mostly used in industry. However, *ITO* is faced with substantial problems due to high demand, high cost and scarcity of indium. To counter the above mentioned threats, the European industry requires the replacement of *ITO* with lower cost and readily available metals [15]. Since doped zinc-oxide, that is proposed as potential replacements for *ITO*, has a much lower conductivity [16], lately, lanthanum-doped $BaSnO_3$ (*BSO*) thin films have been used due to its high conductivity ($\sim 10^4$ S/cm) and high mobility ($183\text{ cm}^2/\text{Vs}$) [17–22]. While the studies on the usage of other materials in place of indium continue, researchers still work on indium based devices such as solar cells [23], light emitting diodes [24], liquid crystal displays [25], gas sensors [26] and ultraviolet photo detectors [27]. Despite the impressive advancement in In_2O_3 thin film based optoelectronic devices, many material properties of In_2O_3 thin films have not been yet well understood [28]. For example, there have not been detailed study on the effect of the infrared lights to the In_2O_3 thin film based optoelectronic devices. There are only a few reports on the infrared light- In_2O_3 thin film interaction such as the near infrared (*NIR*) transparency of molybdenum doped In_2O_3 thin films [29] and infrared emissivity of un-doped In_2O_3 nanoparticles [30]. Unfortunately, these precious studies are only based on the In_2O_3 thin films. To the best of our knowledge, there have not been a study on the In_2O_3 based metal-semiconductor-metal (*MSM*) infrared (*IR*) photodetectors with the effect of *Pt* noble metal doping.

Therefore, in this study, analyses on the structural, morphological and electrical properties of In_2O_3 thin films were carried out for the investigation of *Pt* noble metal doping effect. For that purpose, the *Pt* doping amount was arranged via the changing sputter power as 0W (un-doped), 11W and 14W. The dislocation density (δ), strain (ϵ) values, and grain size (D), which can be identified as structural parameters, were obtained by means of X-Ray Diffraction (*XRD*) measurements. Morphological characteristics of the In_2O_3 thin films were achieved by using atomic force microscopy (*AFM*) measurement system. The fabrication of In_2O_3 *MSM IR* photodetectors were realized via lithographic processes. The dark Current-Voltage (*I–V*) measurements of In_2O_3 *MSM IR* photodetectors were further determined at room temperature. Subsequently, the saturation current (I_0), Schottky barrier height (Φ_B), series resistance (R_s), and ideality factor (n) called as the electrical parameters were extracted from dark *I–V* characteristics. Moreover, the carrier lifetime (τ) and diffusion length (L) in *MSM IR* devices were calculated using the experimental mobility (μ) and carrier density (N) data obtained by Hall Effect measurements at constant magnetic field (0.5 T) and temperature (300 K).

2. EXPERIMENTAL DETAILS

The In_2O_3 thin films with the thickness of approximately 100 nm were deposited on the n type Si substrates by using high purity (99.999%) Pt and In_2O_3 targets at the constant pressure ($4.4 \times 10^{-3}\text{ mbar}$) and temperature ($23\text{ }^\circ\text{C}$). The Pt sputter power was varied to be 0 W (undoped), 11 W , and 14 W so that the Pt doping effect on the properties of the In_2O_3 thin films was investigated. The other deposition parameters, which are In_2O_3 sputter power, deposition temperature, deposition pressure, argon flow and sample-target distance, given in Table-I were kept constant at whole deposition processes. Therefore, the structures called as $InOx-1$, $InOx-2$, and $InOx-3$ were produced for their characterizations and fabrications. Then, X-Ray Diffraction (XRD), and Atomic Force Microscopy (AFM) measurements were taken for obtaining the structural and morphological properties of the In_2O_3 thin films.

Table. I. The deposition parameters of the $In_2O_3/n\text{-Si}$ structures.

Sample	Structure	Sputter Power		Deposition			Sample-Target Distance (mm)
		In_2O_3 (W)	Pt (W)	Temperature ($^\circ\text{C}$)	Pressure (mtorr)	Thickness (nm)	
<i>InOx-1</i>	$In_2O_3/n\text{-Si}$	100	0	23	30	99	120
<i>InOx-2</i>	$In_2O_3/n\text{-Si}$	100	11	23	30	101	120
<i>InOx-3</i>	$In_2O_3/n\text{-Si}$	100	14	23	30	100	120

The samples were fabricated by using the lithographic processes, each of for obtaining the electrical characterizations of the In_2O_3 MSM IR photodetectors. The fabrication included two intertwined Schottky interdigital electrodes, which were placed on the In_2O_3 thin films. The electrode areas with $100\text{ }\mu\text{m}$ line width, $100\text{ }\mu\text{m}$ spacing and $800\text{ }\mu\text{m}$ length were formed by using a mask-aligner system with a special designed photolithographic mask. After forming the interdigital electrode spaces, the high purity (99.999%) Ni (20 nm) and $AuNiGe$ (200 nm) metals were evaporated at room temperature using a thermal evaporation system. Thus, the In_2O_3 MSM IR photodetectors were produced, as schematized Figure. I. The I - V characteristics of these photodetectors were obtained by using a semiconductor parameter analyzing system. Furthermore, the mobility and carrier density values of the devices measured by Hall Effect system at constant magnetic field (0.5 T) and temperature (300 K) were used so that the carrier lifetime (τ) and diffusion length (L) in the MSM IR photodetectors were calculated.

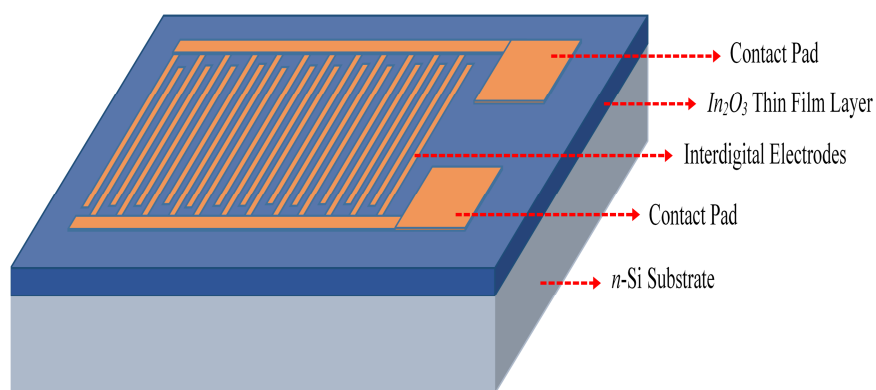


Fig. I. The schematic representation of the In_2O_3 MSM IR photodetector.

3. RESULTS and DISCUSSION

The performance of a MSM IR photodetector is crucially dependent on some important characteristics such as morphological, structural, and particularly electrical [31–33].

3.1. Structural Characteristics of the In_2O_3 Thin Films

The recorded ω - 2θ XRD spectra of the In_2O_3 thin films were presented in *Figure. II*. It can be noticed from the *Figure. II* that, the high intense diffraction peak arises from *n-Si* substrate [34] as the other diffraction peaks belong to the body-centered cubic structure of In_2O_3 from JCPDS Card No. 6 - 416.

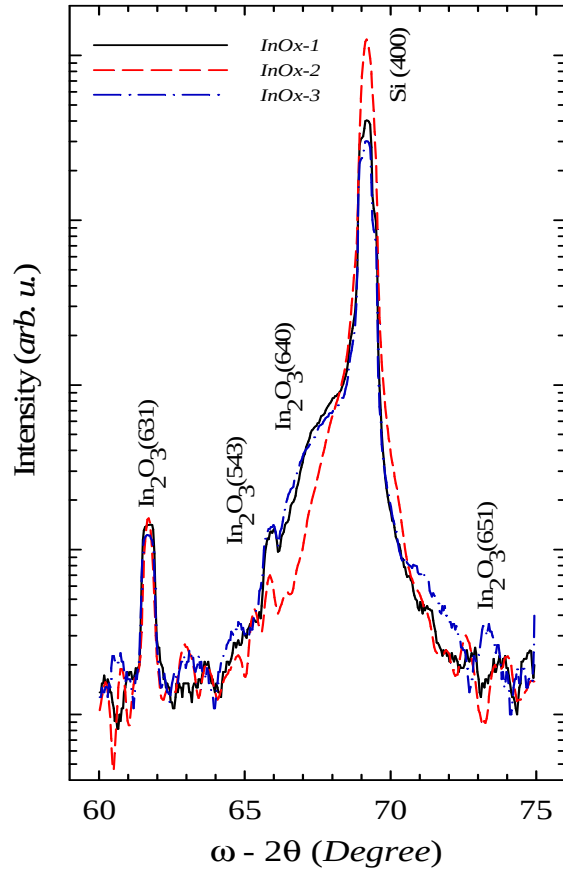


Fig. II. The ω -2 θ XRD spectra of In_2O_3 thin films.

The reflection peaks occurring from the different planes such as Si (400), In_2O_3 (631), In_2O_3 (543), In_2O_3 (640), and In_2O_3 (651) peaks mean that these thin films have polycrystalline structure, as shown in *Figure II*.

In addition, the grain size (D), strain (ε), dislocation density (δ), lattice parameter (a) and lattice spacing (d) values of In_2O_3 thin films were calculated and given in *Table-II*. The values were obtained from the equations given by;

$$D = \frac{0.9\lambda}{\beta \cos \theta} \quad (1)$$

$$\varepsilon = \frac{1}{\tan \theta} \left(\frac{\lambda}{D \cos \theta} - \beta \right) \quad (2)$$

$$\delta = \frac{1}{D^2} \quad (3)$$

$$d = \frac{\lambda}{2 \sin \theta} \quad (4)$$

$$\frac{1}{d^2} = \frac{h^2 + k^2 + l^2}{a^2} \quad (5)$$

where the peak angle value in the ω - 2θ scan is denoted by θ . Additionally, β , λ and hkl denote the full width at half maximum (*FWHM*) value, the X-Ray wavelength (1.540 Å), and the Miller indices of the In_2O_3 (631) peaks, respectively. [12].

Table. II. The structural parameters of the In_2O_3 samples.

Sample	2θ (deg)	β (deg)	D (nm)	δ ($\times 10^9 \text{ cm}^{-2}$)	ε ($\times 10^{-4}$)	d (Å)	a (Å)
<i>InOx-1</i>	61.660	0.453	20.386	2.406	14.745	1.503	11.930
<i>InOx-2</i>	61.990	0.324	28.478	1.232	10.550	1.502	11.924
<i>InOx-3</i>	61.660	0.458	20.173	2.457	14.901	1.503	11.930

It is seen from *Table-II* that the *InOx-2* thin film has the highest D and lowest δ and ε . Thus, it can be said it has better crystallinity compared to the other samples. Besides, the *FWHM* values of reflection peaks are susceptible to the crystal quality of the handled polycrystalline thin films [35]. So, it can be thought that the better crystal properties of the *InOx-2* thin film may affect to the electrical results of that *MSM IR* device.

The *FWHM*, as shown in *Figure. III*, firstly decreased and then increased with the increasing of *Pt* doping amount. It is also considered that the crystallinity of the *InOx-2* thin film improved because of the decreased both *FWHM* and the dislocation density together with the increased grain size. The similar behavior in “*Pt doped SnO₂ thin film*” was observed in the literature [11].

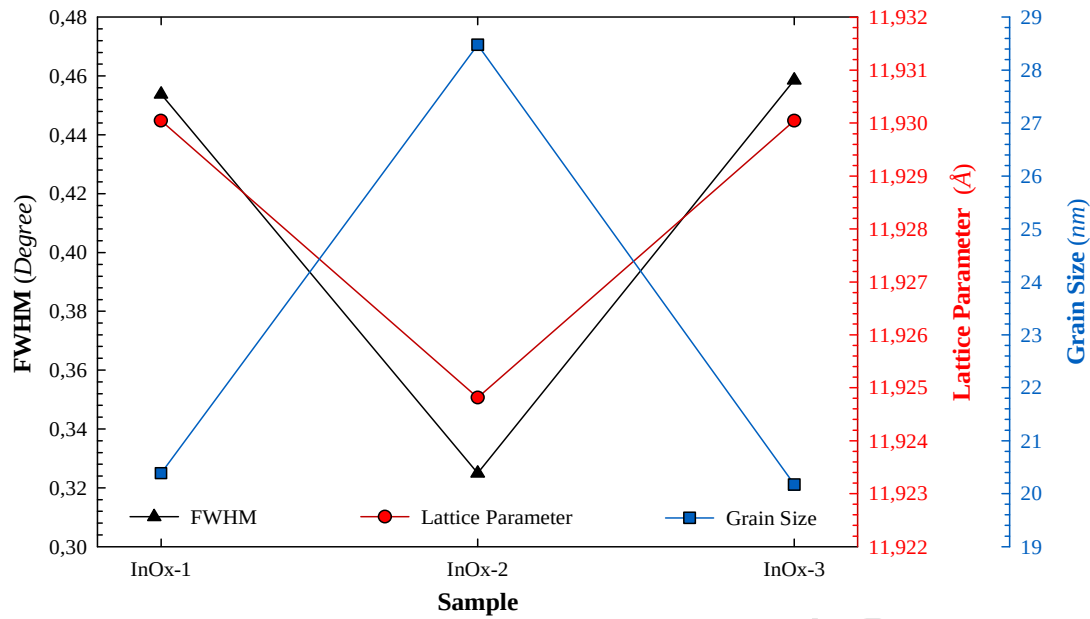


Fig. III. The calculated *FWHM*, lattice parameter and grain size values for the In_2O_3 thin films.

3.2. Morphological Characteristics of the In_2O_3 Thin Films

One of the important factors affecting the operational performance of *MSM IR* photodetectors is surface morphology. The surface characterization methods such as *AFM* can be used to determine the surface characteristics and grain sizes [12]. *Figure. IV* shows three-dimensional (3D) *AFM* images of the thin films with the $3 \times 3 \mu m^2$ scanning area. The grain sizes were evaluated to be 60.79 nm, 71.56 nm, and 46.14 nm from the *AFM* images of *InOx-1*, *InOx-2*, and *InOx-3*, respectively. In addition, the root-mean square (*RMS*) values were obtained to be 2.02 nm, 3.83 nm, and 3.30 nm from the surfaces of the *InOx-1*, *InOx-2*, and *InOx-3*, respectively. If the grain size and *RMS* values listed in *Table-III* are considered, it can be said that they are firstly increasing and then decreasing with the increasing *Pt* sputter power. It means that the crystallinity of the thin films is changed by the *Pt* sputter power. The grain size and *RMS* values of *InOx-2* thin film are bigger than the others. Therefore, it can be said that the changes on the roughness and grain size associate with the performance of the In_2O_3 *MSM IR* photodetectors, especially *InOx-2* based.

Table. III. The grain size and *RMS* values of the In_2O_3 thin films.

Sample	Grain Size (nm)		<i>RMS</i> (nm)	<i>Pt</i> Sputter Power (W)
	<i>XRD Results</i>	<i>AFM Results</i>		
<i>InOx-1</i>	20.38	60.79	2.02	0
<i>InOx-2</i>	28.47	71.56	3.83	11

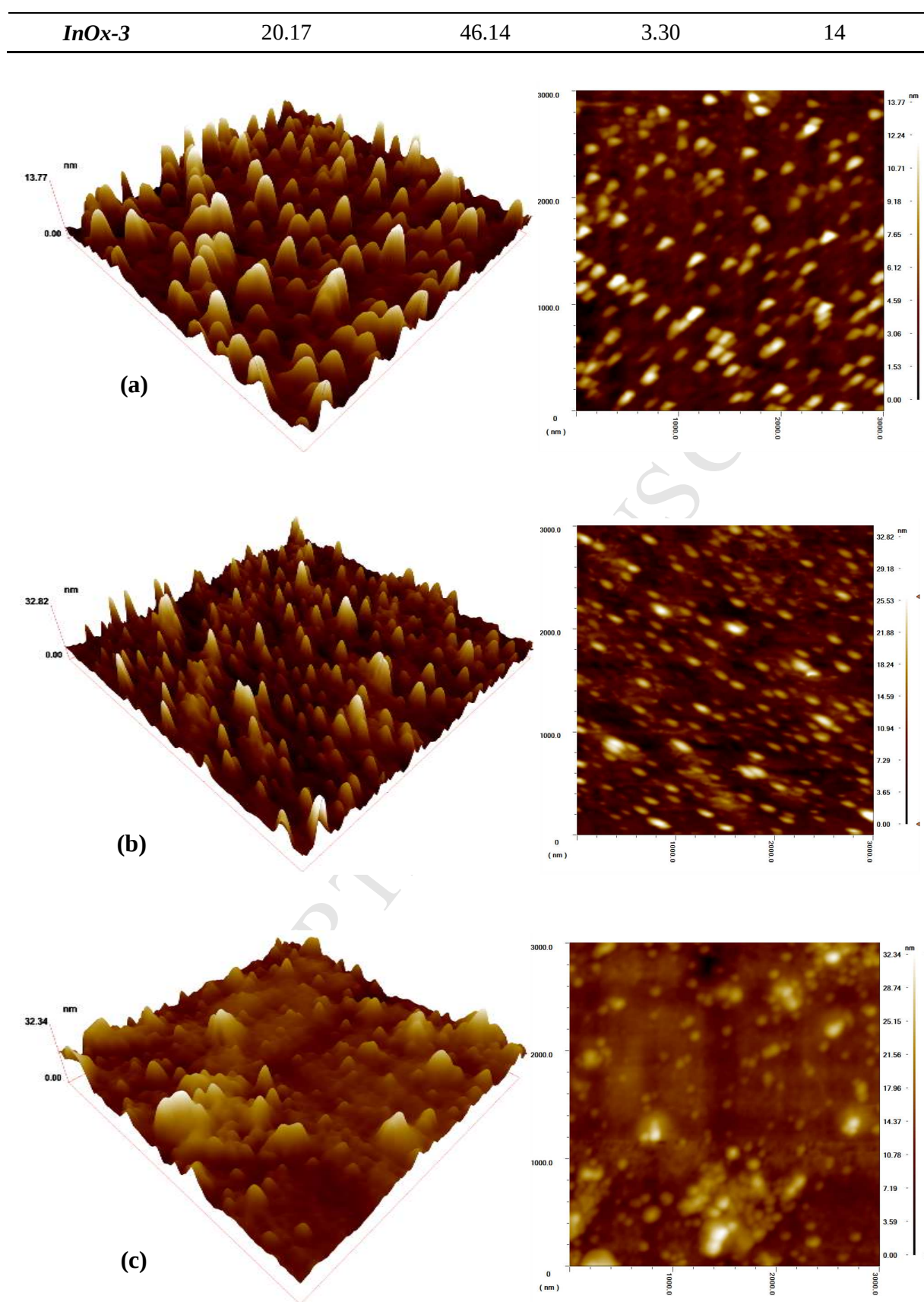


Fig. IV. 2D and 3D AFM images of the In_2O_3 thin films. (a) $InOx-1$, (b) $InOx-2$ and (c) $InOx-3$.

3.3. Electrical Characteristics of the In_2O_3 MSM IR Photodetectors

The reverse and forward bias dark I - V characteristics of In_2O_3 MSM IR photodetectors were determined so that the electrical properties such as the saturation current (I_0), Schottky barrier height (Φ_B), series resistance (R_s), and ideality factor (n) were investigated. Figure. V shows the semi-logarithmic $\ln I$ - V characteristics of the In_2O_3 MSM IR photodetectors at room temperature.

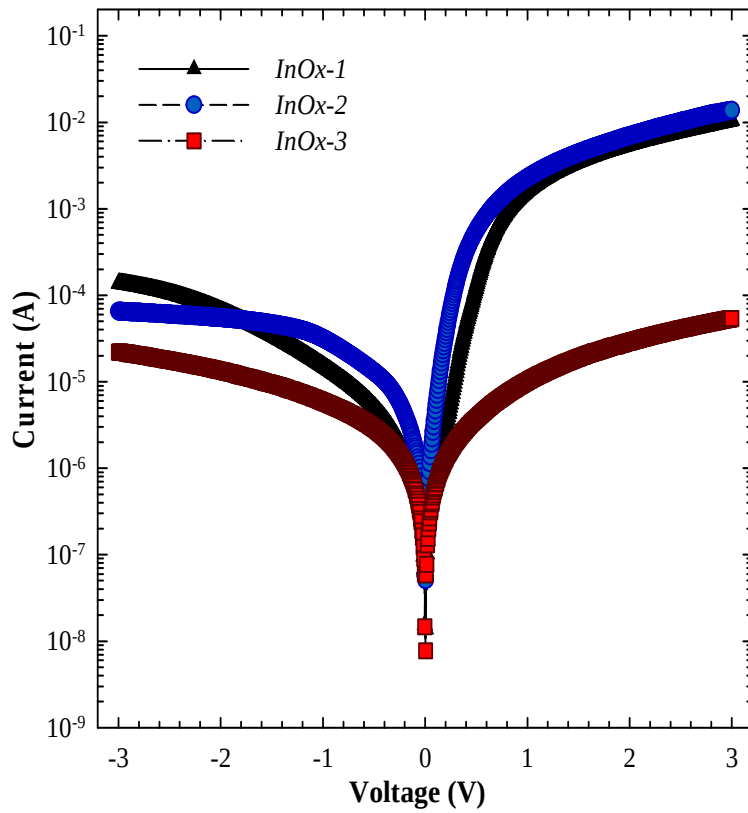


Fig. V. The $\ln I$ - V characteristics of In_2O_3 MSM IR photodetectors.

In order to quantitatively analyze the In_2O_3 MSM IR photodetectors, the standard thermionic emission (TE) theory was used as follows [36]:

$$I = I_o \exp\left(\frac{q(V - IR_s)}{nkT}\right) \quad (6)$$

where I_o , q , n , T , and k denote the saturation current, electronic charge, ideality factor, temperature and Boltzmann's constant, respectively. The voltage drop across the series resistance of the diode is given by the term of IR_s .

I_o can be extracted from the straight line intercept of $\ln I$ - V plot at zero bias voltage [37]. Thus, it is expected as:

$$I_o = AA^*T^2 \exp\left(-\frac{q\Phi_B}{kT}\right) \quad (7)$$

where A , A^* , and Φ_B , respectively, denote the Schottky contact area, effective Richardson constant, and effective barrier height at zero bias. The A^* of In_2O_3 is given by,

$$A^* = 120\alpha \left(\frac{m_e}{m_0}\right) (Acm^{-2}K^{-2}) \quad (8)$$

where α is an empirical factor on the order of unity, m_e is the effective mass and equals $0.3m_0$. We can assume that the empirical factor is 1 [38]. Thus, it can be used as $36 Acm^{-2}K^{-2}$ for In_2O_3 .

The Φ_B was derived from the extrapolated I_o and was given by the relation:

$$\Phi_B = \frac{kT}{q} \ln\left(\frac{AA^*T^2}{I_o}\right) \quad (9)$$

Furthermore, n derived from the slope of linear region of the forward-bias $\ln I$ - V plot [39,40] is given by;

$$n = \frac{q}{kT} \left(\frac{dV}{d(\ln I)}\right) \quad (10)$$

The similar $\ln I$ - V behavior, as seen from the *Figure. V*, was observed due to the effects of the interfacial layers, interface states, and series resistance [41]. Additionally, an interfacial layer emerged between metal and semiconductor may cause the higher values of ideality factor. An interfacial layer may be occurred in both surface preparation and fabrication processes [42].

The experimental values of n and Φ_B were obtained from Eq. (9) and Eq. (10). The values of n and Φ_B given in Table-IV are 3.08, 1.85 and 4.55; 0.69 eV, 0.67 eV and 0.69 eV for *InOx-1*, *InOx-2* and *InOx-3 In₂O₃ MSM IR photodetectors*, respectively.

The Φ_B , n , and R_s were also obtained by using *Cheung and Cheung* functions [43]. These functions were given as:

$$\frac{dV}{d(\ln I)} = n \left(\frac{kT}{q} \right) + IR_s \quad (11)$$

$$H(I) = V - n \left(\frac{kT}{q} \right) \ln \left(\frac{I}{AA * T^2} \right) = IR_s + n\Phi_B \quad (12)$$

The experimental $dV/d(\ln I)$ - I and $H(I)$ - I plots for each of the devices at room temperature can be seen in *Figure. VI* and *Figure. VII*. The plots exhibit a straight line for the data of the downward curvature region in the forward bias I - V . So, the R_s value is found from the $dV/d(\ln I)$ - I plot's slope while the y-axis intercept of this plot gives the n value. The values of Φ_B and R_s can be also determined from $H(I)$ - I plot using the same method.

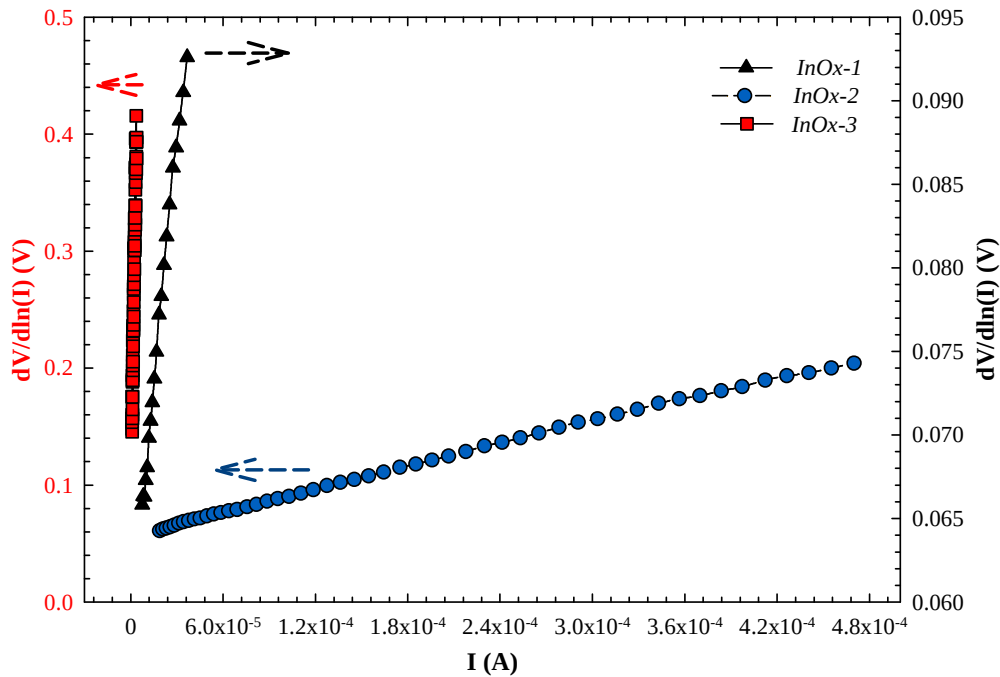


Fig. VI. The experimental $dV/d\ln(I)$ vs I plot of In_2O_3 MSM IR photodetectors.

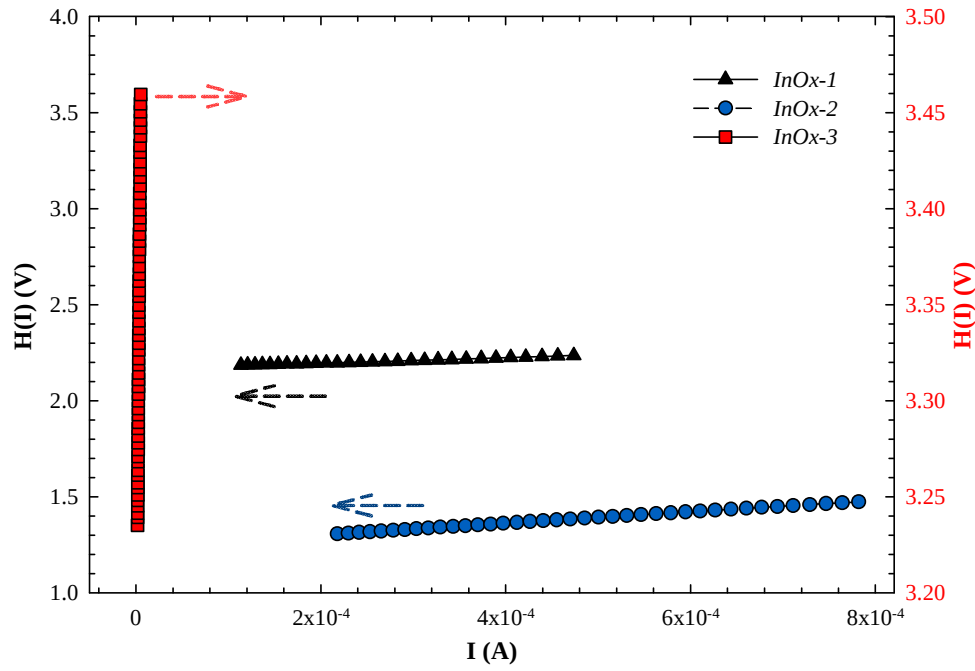


Fig. VII. The experimental $H(I)$ vs I plot of In_2O_3 MSM IR photodetectors.

Thus, the values of n and R_s were respectively found to be 963.13Ω , 321.06Ω and 86668.00Ω ; 2.26, 2.23 and 3.54 for $InOx-1$, $InOx-2$ and $InOx-3$ MSM IR photodetectors from $dV/d(\ln I)$ - I plot. The values of Φ_B and R_s were respectively found to be 138.15Ω , 297.54Ω and 70121.00Ω ; 0.70 eV , 0.67 eV and 0.68 eV for $InOx-1$, $InOx-2$ and $InOx-3$ In_2O_3 MSM IR photodetectors from $H(I)$ - I plot. The obtained values were also given in Table-IV.

In addition, Norde method modified by Lien was used so that Φ_B and R_s were determined [44,45]. The $F(V, \gamma)$ function was defined using this method as:

$$F(V, \gamma) = \frac{V}{\gamma} - \frac{q}{kT} \ln \left(\frac{I(V)}{AA * T^2} \right) \quad (13)$$

where γ is a dimensionless integer being greater as compared to the ideality factor of device, and $I(V)$ is current value derived from I - V curve. Thus, the graph of voltage dependent $F(V, \gamma)$ function was drawn and was given in Figure. VIII.

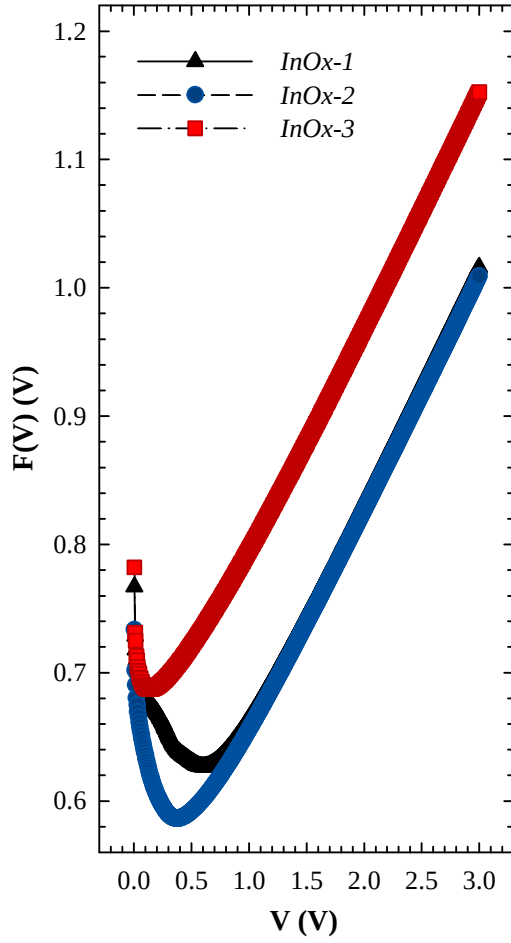


Fig. VIII. The $F(V)$ vs V plot of In_2O_3 MSM IR photodetectors.

The Φ_B and R_s values given in Eqs. (14) and (15) can be derived by determining the minimum point of the $F(V)$ - V plot.

$$\Phi_B = F(V_{min}) + \frac{V_{min}}{\gamma} - \frac{kT}{q} \quad (14)$$

$$R_s = \frac{kT}{qI_{min}}(\gamma - n)$$

(15)

The coordinates of minimum point in the $F(V)$ - V plot are denoted with V_{min} and $F(V_{min})$, the forward bias current value corresponding to V_{min} is I_{min} [45].

The values of Φ_B obtained from Eq. (14) were found to be 0.72 eV, 0.64 eV and 0.69 eV for *InOx-1*, *InOx-2* and *InOx-3* MSM devices, respectively. The obtained R_s values given in Table-IV were determined from Eq. (15) to be 159.32 Ω , 293.04 Ω and 14436.69 Ω for *InOx-1*, *InOx-2* and *InOx-3* MSM IR photodetectors, respectively.

Table. IV. The electrical parameters for In_2O_3 MSM IR photodetectors.

<i>Parameters</i>	<i>Method</i>	<i>InOx-1</i>	<i>InOx-2</i>	<i>InOx-3</i>
I_o (μA)	$\ln(I)-V$	23.52	58.60	25.30
n	$\ln(I)-V$	3.08	1.85	4.55
	$dV-d\ln(I)$	2.26	2.23	3.54
Φ_B (eV)	$\ln(I)-V$	0.69	0.67	0.69
	$H(I)$	0.70	0.67	0.68
	$F(V)$	0.72	0.64	0.69
R_s (Ω)	$dV-d\ln(I)$	963.13	321.06	86668.00
	$H(I)$	138.15	297.54	70121.00
	$F(V)$	159.32	293.04	14436.69

As seen from the Table-IV, the Φ_B values of *InOx-1*, *InOx-2* and *InOx-3* MSM IR photodetectors were approximately found to be 0.70 eV, 0.67 eV and 0.69 eV, respectively. These values correspond the infrared wavelengths of 1.771 μm , 1.851 μm and 1.797 μm . Thus, it can be said that all indium oxide MSM IR devices can work as an infrared photodetector at their infrared wavelengths.

On the other hand, it can be seen from Table-IV, there is a Pt doping limit for the indium oxide thin films. When the indium oxide was doped with Pt, which has 11W sputter power, the electrical behaviors such as n , Φ_B , and R_s values were improved. However, when the Pt sputter power was increased from 11W to 14W, the electrical behaviors were getting worse. The n , Φ_B and R_s were bigger than that of the other MSM IR devices. Additionally, the *InOx-2* MSM IR photodetector has a lower ideality factor being a way of measuring how the diode accurately follows the ideal diode behaviors (like $n=1$) than the other MSM IR photodetectors. The *InOx-2* MSM IR photodetector has also a higher saturation current as compared to that of *InOx-1* and *InOx-3* MSM IR photodetectors. Namely, the MSM IR photodetector with larger indium oxide grain size shows a higher saturation current than that of both MSMs with smaller indium oxide grain size. Thus, it can be said that, the size of indium oxide grains plays a crucial role on the operating of MSM IR photodetectors. The similar results with the same electrode design were exhaustively discussed in our previous MSM IR photodetector paper [12]. Additionally, the optimizing of semiconductor layer causes a change in the saturation current. The dark current results of MSMs given in literature also show that the semiconductor beneath the metal electrodes can be affected the carriers' flow because of two interfaces to

traverse across, *Metal/Semiconductor* interface and *Semiconductor/Si* interface [12,46]. That reason can be used to explain why *InOx-2 MSM IR* photodetector has higher saturation current than the others. Therefore, it can be said that *InOx-2 MSM IR* photodetector with larger indium oxide grains has less rooms for carriers therefore resulting in an increment on the saturation current.

Moreover, the electro-optical performance of an *In₂O₃ MSM IR* photodetector is dependent on some important parameters such as mobility (μ), carrier density (N), diffusion coefficient (D), carrier lifetime (τ), and diffusion length (L). Thus, the N and μ were calculated from the Hall coefficient (R_H) measured by using Hall Effect system at constant magnetic field (0.5 T) and temperature (300 K). The D , τ and L values for *In₂O₃ MSM IR* photodetectors were calculated using the obtained N and μ data. The obtained values are given in *Table-V*.

The total recombination lifetime, τ is given as [47];

$$\frac{1}{\tau} = \frac{1}{\tau_{SRH}} + \frac{1}{\tau_R} + \frac{1}{\tau_A}. \quad (16)$$

In terms of doping concentrations N , the *Eq. (16)* can be rewritten as;

$$\tau = \left[BN + CN^2 + \tau_{SRH}^{-1} \right]^{-1}. \quad (17)$$

It can be obviously seen from the *Eq. (16)* and *Eq. (17)*, the total recombination lifetime related to the N separates into the three recombination parts as radiative, Auger and Shockley–Read–Hall (*SRH*) [31]. From higher doping level than 10^{18} cm^{-3} , Auger recombination dominates the other mechanisms. Therefore, the total recombination lifetime is related to the inverse square of the doping concentration [48,49]. As a result of this limitation, total recombination lifetime is given as [12];

$$\tau = \tau_A \Rightarrow \tau = \left[CN^2 \right]^{-1}. \quad (18)$$

Here, the Auger recombination coefficients (C) are $5.6 \times 10^{-31} \text{ cm}^6 \text{ s}^{-1}$ for *In₂O₃* [50] and $3.8 \times 10^{-31} \text{ cm}^6 \text{ s}^{-1}$ for *Si* [51].

The other important parameter for an *MSM IR* photodetector is the diffusion length (L). L can be derived from the τ as [31];

$$L_{n,p} = \sqrt{\tau D_{n,p}}, \quad (19)$$

where the diffusion coefficient D is given by the Einstein relation as;

$$D = \mu \left(\frac{kT}{q} \right). \quad (20)$$

Here, μ is the carrier mobility, k is the Boltzmann's constant and q is the electrical charge of electron [52].

Table.V. The electro-optical results of the In_2O_3 *MSM IR* photodetectors.

<i>Device</i>	<i>T</i> (K)	<i>B</i> (T)	<i>N</i> ($10^{18}cm^{-3}$)	μ ($cm^2/V.s$)	R_{Bulk} ($\Omega.cm$)	<i>D</i> (cm^2/s)	τ (ns)	<i>L</i> (μm)
<i>InOx-1</i>	300.633	0.5	1.093	41.257	0.0628	1.069	1492.3	12.629
<i>InOx-2</i>	300.257	0.5	3.817	69.764	0.0089	1.805	122.5	4.703
<i>InOx-3</i>	300.468	0.5	2.546	33.126	0.0277	0.858	275.4	4.861

Consequent of the semi-empirical calculations as seen the *Table-V*; it can be said that, the electro-optical output parameters such as μ , N , D , τ and L were also affected by the *Pt* doping. Besides, there is a similar *Pt* doping situation obtained from *I-V*. When the In_2O_3 was doped with *11W Pt*, the electro-optical output values of *MSM IR* photodetector were improved. Contrary, when the *Pt* sputter power was increased from *11W* to *14W*, the electro-optical behaviors were getting worse. The μ , N and D values of the *InOx-2 MSM IR* photodetector are higher than the other In_2O_3 *MSM IR* photodetectors. Additionally, the τ and L values of the *InOx-2 MSM IR* photodetector are the lowest ones.

4. CONCLUSIONS

In order to investigate *Pt* doping effect on room temperature fabricated In_2O_3 based *MSM IR* photodetectors, the *Pt* sputter power was arranged to be *0W* (un-doped), *11W* and *14W* by

using confocal magnetron sputtering system. The experimental and theoretical evidences proved that the *Pt* doping significantly affected the structural, morphological, and electrical properties of room temperature fabricated In_2O_3 MSM IR photodetectors. Especially, the *InOx-2* type structure which was deposited at 100W In_2O_3 and 11W *Pt* sputter powers at room temperature stands out in In_2O_3 based MSM IR photodetectors. In addition, this work supplies a contribution to the insufficient In_2O_3 MSM IR photodetector literature with easy fabrication processes and low production costs.

Acknowledgements: This work is supported by the Ministry of Development of Turkey under project numbers: 2011K120290 and 2016K121220.

References

- [1] C.G. Granqvist, Transparent conductive electrodes for electrochromic devices: A review, *Appl. Phys. A Solids Surfaces*. 57 (1993) 19–24. doi:10.1007/BF00331211.
- [2] D. Gogova, A. Iossifova, T. Ivanova, Z. Dimitrova, K. Gesheva, Electrochromic behavior in CVD grown tungsten oxide films, *J. Cryst. Growth*. 198–199 (1999) 1230–1234. doi:10.1016/S0022-0248(98)01051-3.
- [3] Y. Şafak Asar, T. Asar, Ş. Altındal, S. Özçelik, Dielectric spectroscopy studies and ac electrical conductivity on $(\text{AuZn})/\text{TiO}_2/\text{p-GaAs}(110)$ MIS structures, *Philos. Mag.* 95 (2015) 2885–2898. doi:10.1080/14786435.2015.1081301.
- [4] B. Coskun, T. Asar, U. Akgul, K. Yildiz, Y. Atici, Investigation of structural and electrical properties of Zirconium dioxide thin films deposited by reactive RF sputtering technique, *Ferroelectrics*. 502 (2016) 147–158. doi:10.1080/00150193.2016.1235453.
- [5] C.G. Granqvist, *Handbook of Inorganic Electrochromic Materials*, 1995. doi:10.1016/B978-044489930-9/50014-3.
- [6] D. Gogova, L.K. Thomas, B. Camin, Comparative study of gasochromic and electrochromic effect in thermally evaporated tungsten oxide thin films, *Thin Solid Films*. 517 (2009) 3326–3331. doi:10.1016/j.tsf.2008.12.020.
- [7] D. Liu, W. Lei, S. Qin, L. Hou, Z. Liu, Q. Cui, Y. Chen, Large-scale synthesis of hexagonal corundum-type In_2O_3 by ball milling with enhanced lithium storage capabilities, *J. Mater. Chem. A*. 1 (2013) 5274–5278. doi:10.1039/c3ta00182b.
- [8] M.A. Islam, M. Nuruzzaman, R.C. Roy, J. Hossain, K.A. Khan, Investigation of Electrical and Optical Transport Properties of N-type Indium Oxide Thin Film, *Am. J. Eng. Res.* 4 (2015) 62–67.
- [9] H.I. Yeom, J.B. Ko, G. Mun, S.H.K. Park, High mobility polycrystalline indium oxide thin-film transistors by means of plasma-enhanced atomic layer deposition, *J. Mater. Chem. C*. 4 (2016) 6873–6880. doi:10.1039/C6TC00580B.
- [10] P.D.C. King, T.D. Veal, F. Fuchs, C.Y. Wang, D.J. Payne, A. Bourlange, H. Zhang, G.R. Bell, V. Cimalla, O. Ambacher, R.G. Egdell, F. Bechstedt, C.F. Mcconville, Band gap, electronic structure, and surface electron accumulation of cubic and rhombohedral In_2O_3 , *Phys. Rev. B*. 79 (2009) 205211. doi:10.1103/PhysRevB.79.205211.
- [11] T. Asar, B. Korkmaz, S. Özçelik, Effect of platinum doping on the structural and electrical properties of SnO_2 thin films, *J. Exp. Nanosci.* 11 (2016) 1285–1306. doi:10.1080/17458080.2016.1212408.
- [12] T. Asar, S. Özçelik, Barrier enhancement of Ge MSM IR photodetector with Ge layer optimization, *Superlattices Microstruct.* 88 (2015) 685–694. doi:10.1016/j.spmi.2015.10.034.
- [13] D. Selvakumar, N. Dharmaraj, N.S. Kumar, V.C. Padaki, Oxygen Sensing Properties of

Platinum Doped Indium Oxide Nanoparticles Prepared by Hydrothermal Method, *Synth. React. Inorganic, Met. Nano-Metal Chem.* 45 (2015) 753–758. doi:10.1080/15533174.2013.843548.

- [14] J. Tu, N. Li, X. Lai, Y. Chi, Y. Zhang, W. Wang, X. Li, J. Li, S. Qiu, H₂S-sensing properties of Pt-doped mesoporous indium oxide, *Appl. Surf. Sci.* 256 (2010) 5051–5055. doi:10.1016/j.apsusc.2010.03.055.
- [15] T.T. Werner, G.M. Mudd, S.M. Jowitt, Indium: key issues in assessing mineral resources and long-term supply from recycling, *Appl. Earth Sci.* 124 (2015) 213–226. doi:10.1179/1743275815Y.0000000007.
- [16] Y. Igasaki, H. Saito, The effects of deposition rate on the structural and electrical properties of ZnO:Al films deposited on (1120) oriented sapphire substrates, *J. Appl. Phys.* 70 (1991) 3613–3619. doi:10.1063/1.349258.
- [17] H. Paik, Z. Chen, E. Lochocki, A. Seidner H., A. Verma, N. Tanen, J. Park, M. Uchida, S. Shang, B.-C. Zhou, M. Brützm, R. Uecker, Z.-K. Liu, D. Jena, K.M. Shen, D.A. Muller, D.G. Schlom, Adsorption-controlled growth of La-doped BaSnO₃ by molecular-beam epitaxy, *APL Mater.* 5 (2017) 116107. doi:10.1063/1.5001839.
- [18] S. Raghavan, T. Schumann, H. Kim, J.Y. Zhang, T.A. Cain, S. Stemmer, High-mobility BaSnO₃ grown by oxide molecular beam epitaxy, *APL Mater.* 4 (2016) 016106. doi:10.1063/1.4939657.
- [19] X. Luo, Y.S. Oh, A. Sirenko, P. Gao, T.A. Tyson, K. Char, S.-W. Cheong, High carrier mobility in transparent Ba_{1-x}La_xSnO₃ crystals with a wide band gap, *Appl. Phys. Lett.* 100 (2012) 172112. doi:10.1063/1.4709415.
- [20] H.J. Kim, U. Kim, H.M. Kim, T.H. Kim, H.S. Mun, B.G. Jeon, K.T. Hong, W.J. Lee, C. Ju, K.H. Kim, K. Char, High mobility in a stable transparent perovskite oxide, *Appl. Phys. Express.* 5 (2012) 061102. doi:10.1143/APEX.5.061102.
- [21] K. Ganguly, P. Ambwani, P. Xu, J.S. Jeong, K.A. Mkhoyan, C. Leighton, B. Jalan, Structure and transport in high pressure oxygen sputter-deposited BaSnO_{3-δ}, *APL Mater.* 3 (2015) 062509. doi:10.1063/1.4919969.
- [22] A. Prakash, P. Xu, A. Faghaninia, S. Shukla, J.W. Ager, C.S. Lo, B. Jalan, Wide bandgap BaSnO₃ films with room temperature conductivity exceeding 10⁴ Scm⁻¹, *Nat. Commun.* 8 (2017) 15167. doi:10.1038/ncomms15167.
- [23] Y.Z. Dawood, M.S. Mohammed, And, A.H. Al-Hamdani, Characterization and Optimization of I₂O₃ Thin Films for Application in Silicon Solar Cells, *J. Mater. Sci. Eng. A.* 2 (2012) 352–356. doi:10.17265/2161-6213/2012.03.009.
- [24] C.C. Wu, C.I. Wu, J.C. Sturm, a. Kahn, Surface modification of indium tin oxide by plasma treatment: An effective method to improve the efficiency, brightness, and reliability of organic light emitting devices, *Appl. Phys. Lett.* 70 (1997) 1348. doi:10.1063/1.118575.
- [25] Y. Shigesato, S. Takaki, T. Haranoh, Electrical and structural properties of low resistivity tin-doped indium oxide films, *J. Appl. Phys.* 71 (1992) 3356. doi:10.1063/1.350931.
- [26] T. Takada, K. Suzuki, M. Nakane, Highly sensitive ozone sensor, *Sensors Actuators B. Chem.* 13 (1993) 404–407. doi:10.1016/0925-4005(93)85412-4.
- [27] D. Zhang, C. Li, S. Han, X. Liu, T. Tang, W. Jin, C. Zhou, Ultraviolet photodetection properties of indium oxide nanowires, *Appl. Phys. A.* 166 (2003) 163–166. doi:10.1007/s00339-003-2099-3.
- [28] D. Beena, K.J. Lethy, R. Vinodkumar, A.P. Detty, V.P. Mahadevanpillai, V. Ganesan, Photoluminescence in laser ablated nanostructured indium oxide thin films, *Optoelectron. Adv. Mater. – RAPID Commun.* 5 (2011) 1–11.
- [29] S. Parthiban, K. Ramamurthi, E. Elangovan, R. Martins, E. Fortunato, Spray deposited molybdenum doped indium oxide thin films with high near infrared transparency and carrier mobility, *Appl. Phys. Lett.* 94 (2009) 212101. doi:10.1063/1.3142424.
- [30] Y.-M. Zhou, Y. Shan, Y.-Q. Sun, H.-X. Ju, Adsorption of collagen to indium oxide nanoparticles and infrared emissivity study thereon, *Mater. Res. Bull.* 43 (2008) 2105–2112. doi:10.1016/j.materresbull.2007.09.007.
- [31] T. Asar, S. Özçelik, E. Özbay, Structural and electrical characterizations of In_xGa_{1-x}As/InP structures for infrared photodetector applications, *J. Appl. Phys.* 115 (2014) 104502. doi:10.1063/1.4868056.

- [32] W.K. Metzger, M.W. Wanlass, R.J. Ellingson, R.K. Ahrenkiel, J.J. Carapella, Auger recombination in low-band-gap n-type InGaAs, *Appl. Phys. Lett.* 79 (2001) 3272–3274. doi:10.1063/1.1418032.
- [33] N. Naderi, M.R. Hashim, Effect of surface morphology on electrical properties of electrochemically-etched porous silicon photodetectors, *Int. J. Electrochem. Sci.* 7 (2012) 11512–11518.
- [34] C. Suryanarayana, M.G. Norton, *X-ray Diffraction: A Practical Approach.*, 1998. doi:10.1017/S143192769800049X.
- [35] S. Maniv, A. Zangvil, Controlled texture of reactively rf-sputtered ZnO thin films, *J. Appl. Phys.* 49 (1978) 2787.
- [36] E.H. Rhoderick, R.H. Williams, *Metal-Semiconductor Contacts*, Second Edi, Clarendon Press, 1988.
- [37] S.M. Sze, *Physics of Semiconductor Devices*, 2nd Edition, 1981.
- [38] L. Qin, P.S. Dutta, S. Sawyer, Photoresponse of indium oxide particulate-based thin films fabricated using milled nanorods grown by the self-catalytic vapor - liquid - solid process, *Semicond. Sci. Technol.* 27 (2012) 045005. doi:10.1088/0268-1242/27/4/045005.
- [39] H.C. Card, E.H. Rhoderick, Studies of tunnel MOS diodes I. Interface effects in silicon Schottky diodes, *J. Phys. D. Appl. Phys.* 4 (1971) 1589–1601. doi:10.1088/0022-3727/4/10/319.
- [40] B. Kinacı, Y. Özen, K. Kızılkaya, T. Asar, S.Ş. Çetin, E. Boyalı, M.K. Öztürk, T. Memmedli, S. Özçelik, Effect of alloy composition on structural, optical and morphological properties and electrical characteristics of $Ga_{x-1}In_xP/GaAs$ structure, *J. Mater. Sci. Mater. Electron.* 24 (2013) 1375–1381. doi:10.1007/s10854-012-0937-9.
- [41] B. Kinaci, T. Asar, S.Ş. Çetin, Y. Özen, K. Kizilkaya, Electrical characterization of Au/ZnO/TiO₂/n-Si and (Ni/Au)/ZnO/TiO₂/n-Si Schottky diodes by using current-voltage measurements, *J. Optoelectron. Adv. Mater.* 14 (2012) 959–963.
- [42] İ. Taşcıoğlu, Ş. Altındal, İ. Polat, E. Bacaksız, The effect of metal work function on the barrier height of metal/CdS/SnO₂/In–Ga structures, *Curr. Appl. Phys.* 13 (2013) 1306–1310. doi:10.1016/j.cap.2013.04.010.
- [43] S.K. Cheung, N.W. Cheung, Extraction of Schottky diode parameters from forward current-voltage characteristics, *Appl. Phys. Lett.* 49 (1986) 85–87. doi:10.1063/1.97359.
- [44] H. Norde, A modified forward I-V plot for Schottky diodes with high series resistance, *J. Appl. Phys.* 50 (1979) 5052–5053. doi:10.1063/1.325607.
- [45] C.D. Lien, F.C.T. So, M.A. Nicolet, An Improved Forward I-V Method for Nonideal Schottky Diodes with High Series Resistance, *IEEE Trans. Electron Devices.* ED-31 (1984) 1502–1503. doi:10.1109/T-ED.1984.21739.
- [46] A. Baharin, M.R. Hashim, Study of electrical characteristics of Ge islands MSM photodetector structure grown on Si substrate using conventional methods, *Semicond. Sci. Technol.* 22 (2007) 905–910. doi:10.1088/0268-1242/22/8/014.
- [47] R.K. Ahrenkiel, R. Ellingson, S. Johnston, M. Wanlass, Recombination lifetime of In_{0.53}Ga_{0.47}As as a function of doping density, *Appl. Phys. Lett.* 72 (1998) 3470–3472. doi:10.1063/1.121669.
- [48] E. Gaubas, J. Vanhellemont, Comparative Study of Carrier Lifetime Dependence on Dopant Concentration in Silicon and Germanium, *J. Electrochem. Soc.* 154 (2007) H231. doi:10.1149/1.2429031.
- [49] Y. Lv, N. Wang, C. Zhuang, P. Li, B. Han, H. Gong, The uniformity of InGaAs in InP/InGaAs/InP by microwave photoconductivity decay (μ -PCD) carrier lifetime measurement, *Semicond. Sci. Technol.* 21 (2006) 771–774. doi:10.1088/0268-1242/21/6/010.
- [50] A. Othonos, M. Zervos, D. Tsokkou, Femtosecond Carrier Dynamics in In₂O₃ Nanocrystals, *Nanoscale Res. Lett.* 4 (2009) 526–531. doi:10.1007/s11671-009-9275-0.
- [51] C.-M. Li, T. Sjödin, H.-L. Dai, Photoexcited carrier diffusion near a Si (111) surface: Non-negligible consequence of carrier-carrier scattering, *Phys. Rev. B.* 56 (1997) 15252.
- [52] E. Kuphal, K. Mause, K. Mieth, A. Eisenbach, F. Fiedler, A. Corbet, Electron diffusion length in InGaAs:Zn derived from heterostructure bipolar transistors, *Solid. State. Electron.* 38 (1995) 795–799. doi:10.1016/0038-1101(94)00182-F.

ACCEPTED MANUSCRIPT

Highlights

- ✓ The undoped and *Pt*-doped In_2O_3 thin films were deposited for *MSM IR* photodetectors.
- ✓ The *Pt* sputter power was regulated as 0W (undoped), 11W and 14W.
- ✓ The *Pt* dopant usage has a prominent effect on properties of *MSM IR* photodetectors.
- ✓ In_2O_3 based *MSM IR* photodetectors are not extensively studied in the literature.